Robustness of Vision Methods under Image Distortions

Image Processing / Vision course project — COCO benchmark

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https://github.com/ShiraTzi/Final-Project-Image-Processing

Design

- Dataset: COCO val2017 — fixed seeded 1,521-image class-coverage-balanced subset; the same paired images across all variants
- 4 tasks / models / metrics: ORB feature matching (match ratio) · YOLOv8n detection (bbox mAP) · Keypoint R-CNN pose (OKS AP) · Panoptic FPN segmentation (PQ)
- 3 distortions × 3 severities: gauss_noise, salt_pepper, motion_blur × low / med / high
- Two repair strategies: classical enhancement (matched per corruption) and fine-tuning the detector

Method / pipeline

- Deterministic per-image corruption: stable seeded RNG; degradation parameters logged per image (incl. motion-blur kernel angle)
- Lossless PNG storage — a JPEG round-trip would reshape the corruption
- Per-image SNR computed against the clean reference
- Matched enhancement per corruption: sigma-adaptive NLM (Gaussian noise), median filter (salt & pepper), non-blind Wiener with the logged kernel (motion blur)
- Fine-tune = per-image mixture of clean + all 9 corruption cells; AdamW lr 1e-4; held-out train split for checkpoint selection
- Fine-tuned model evaluated on clean + distorted + enhanced val cells

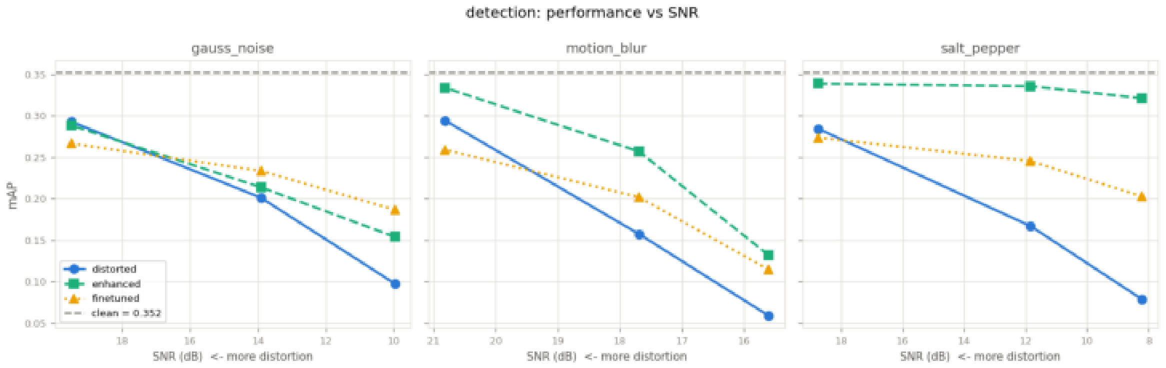
Clean baselines (Phase 1)

Task	Model	Metric	Clean score
Feature matching	ORB	match ratio (vs clean)	1.000
Object detection	yolov8n	bbox mAP@[.5:.95]	0.352
Human pose	Keypoint R-CNN	OKS AP	0.657
Panoptic segmentation	Panoptic FPN	PQ	0.410

- Segmentation PQ splits: things 0.475 / stuff 0.311
- Baselines match the published full-val2017 numbers → the measurement pipeline is validated

Degradation vs SNR (Phase 2)

- Detection mAP falls 0.352 → 0.059 at the worst cell (motion_blur high)
- SNR alone does not predict damage: motion blur has the HIGHEST mean SNR of the three corruptions (gauss_noise 14.5 dB, motion_blur 18.0 dB, salt_pepper 12.9 dB) yet destroys localization tasks most

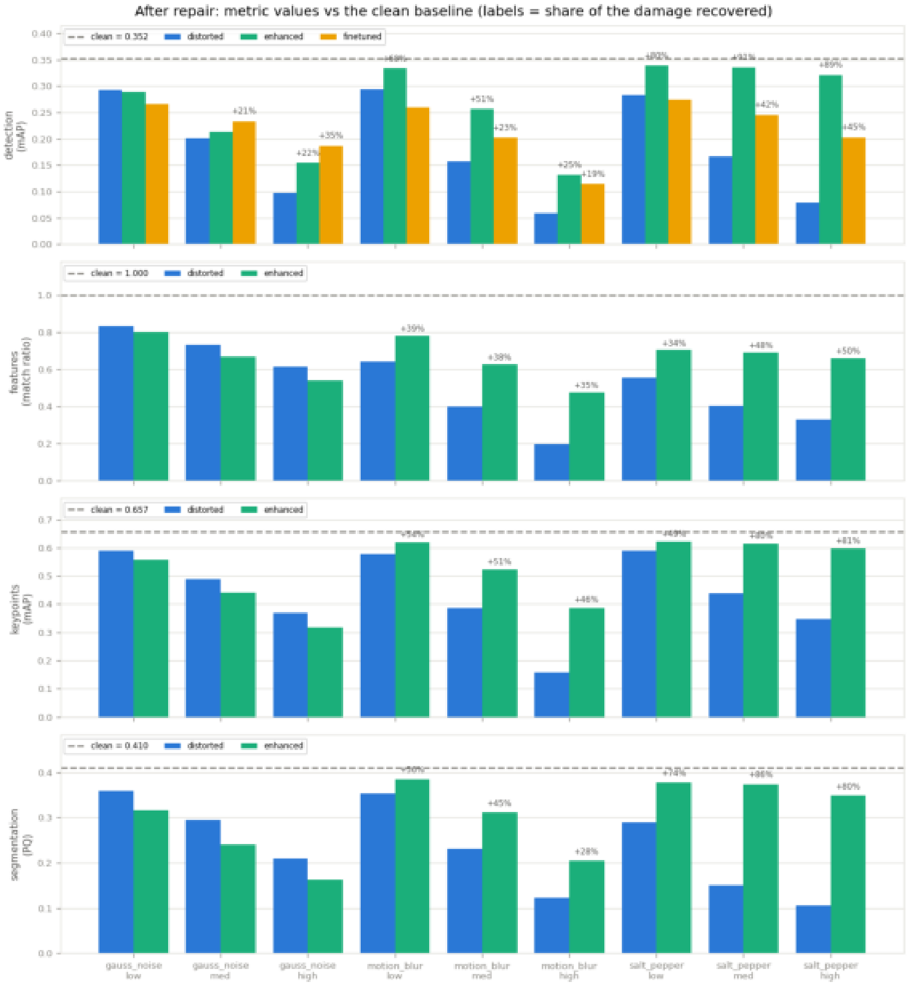


Distortion examples

gauss noise: clean / distorted (high severity) / enhanced (same paired images)

Enhancement (Phase 3): what recovers

- salt pepper: enhancement recovers 45%–88% of the damage across tasks
- gauss noise: enhancement recovers -39%–14% of the damage across tasks
- motion blur: enhancement recovers 37%–48% of the damage across tasks



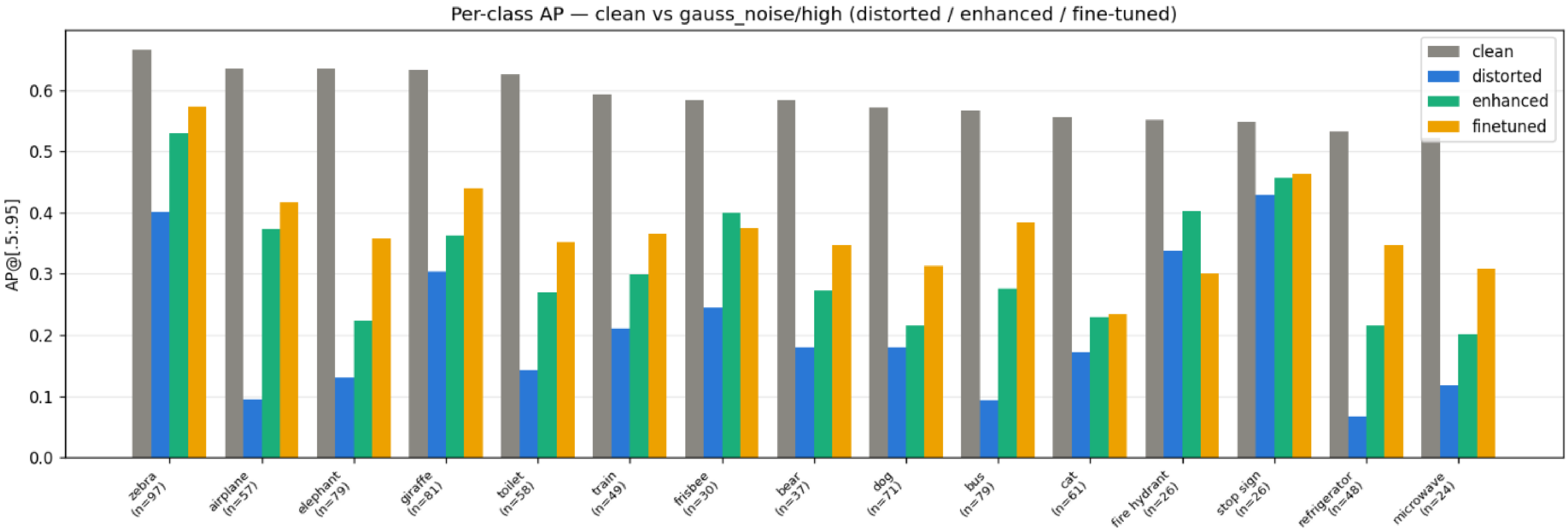
Predictions on images

Gray dashed = ground truth; colored = YOLO predictions (clean / distorted / enhanced / fine-tuned)

Fine-tuning (Phase 4)

- Setup: YOLOv8n fine-tuned on a per-image mixture of clean + all 9 corruption cells (30 epochs, AdamW lr 1e-4)
- On clean images: fine-tuned mAP 0.284 vs pretrained 0.352 — clean cost = 0.068
- In-domain recovery at high severity (absolute mAP): gauss noise +0.089, motion blur +0.056, salt pepper +0.124
- Combined repair (enhance + fine-tune) beats both single repairs on: motion_blur/high

Per-class view



Per-class AP under gauss_noise (high): clean vs distorted vs enhanced vs fine-tuned

Decision matrix / conclusions

- gauss noise: fine-tuning wins (enhance +0.056 vs fine-tune +0.089 mAP at high severity)
- salt pepper: classical enhancement wins (enhance +0.242 vs fine-tune +0.124 mAP at high severity)
- motion blur: classical enhancement wins (enhance +0.073 vs fine-tune +0.056 mAP at high severity)
- Enhancement is corruption-specific — each corruption needs its matched restorer
- Fine-tuning needs mixture training — single-corruption training negative-transfers to every other cell
- Blur is only repairable because the kernel is known (non-blind Wiener)

Limitations

- ORB match ratio structurally penalizes smoothing enhancers — it is a fidelity-to-clean metric, not a task-utility metric
- Wiener deconvolution uses the logged (oracle) kernel; blind deconvolution with a wrong angle measured WORSE than no restoration
- yolov8n is the smallest detector in its family — absolute mAP is modest
- Evaluation subset is ≈1.5k images (class-coverage-balanced, seeded)